

Hand-in 1

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1. (a) False
(b) False
(c) False
(d) True
2. Consider $f : \mathbb{N} \rightarrow A$ where $f(n) = \pi n$. Now to prove this is a bijection, we must prove injectivity and surjectivity.
Injectivity: Suppose $f(n_1) = f(n_2)$ for any $n_1, n_2 \in \mathbb{N}$, then, by the function definition, we have that $\pi n_1 = \pi n_2 \Rightarrow n_1 = n_2$ and thus f is an injection.
Surjectivity: Now suppose $y \in A$, then by the definition of A , $\exists n \in \mathbb{N}$ such that $y = \pi n$. Then $f(n) = \pi n = y$ and thus f is surjective.
 f is now injective and surjective so f is a bijection, thus $|\mathbb{N}| = |A|$
3. Note that the set of irrational numbers can be written as $\{x \in \mathbb{R} : x \notin \mathbb{Q}\} = \mathbb{R} - \mathbb{Q}$. Now suppose that the set of irrational numbers is countable. Since $\mathbb{Q}$ is countable, we know that $\mathbb{Q} \cup (\mathbb{R} - \mathbb{Q})$ is also countable (the union of countable sets is countable). But notice that $\mathbb{Q} \cup (\mathbb{R} - \mathbb{Q}) = \mathbb{R}$. Thus we have that $\mathbb{R}$ is countable, but this is a contradiction and so the initial claim that the set of irrational numbers is countable is false, and so the set of irrational numbers is uncountable